

# **Demand for Labour in Competitive Labour Markets**

ECON 300 - Labour Economics

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Winter 2026

# Learning Objectives

- Understand how firms decide how much labour to employ to produce a given level of output.
- Distinguish **short run** (some inputs fixed) from **long run** (all inputs variable).
- See why labour demand is **downward sloping** with respect to the wage.
- Learn the determinants of the **elasticity of labour demand**.
- Analyze impacts of **free trade**, globalization, offshoring, and ICT on labour demand.
- Compare implications under **perfect competition** vs. **monopoly** in product markets.

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# Main Questions

- Is there empirical support for downward-sloping labour demand?
- How competitive is Canadian labour globally? How have productivity and wages evolved?
- What is the impact of the Internet and technological change on labour demand?
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- Two inputs: Labour  $N$  and capital  $K$ ; output  $Q = F(K, N)$ .
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# Competitive Firm: Properties

- Price taker in output and labour markets.
- Can hire labour without affecting the market wage;  $MC = AC = w$ .
- Hire until  $MRP = w$ ; the **short-run labour demand** is the  $MRP_N$  curve.

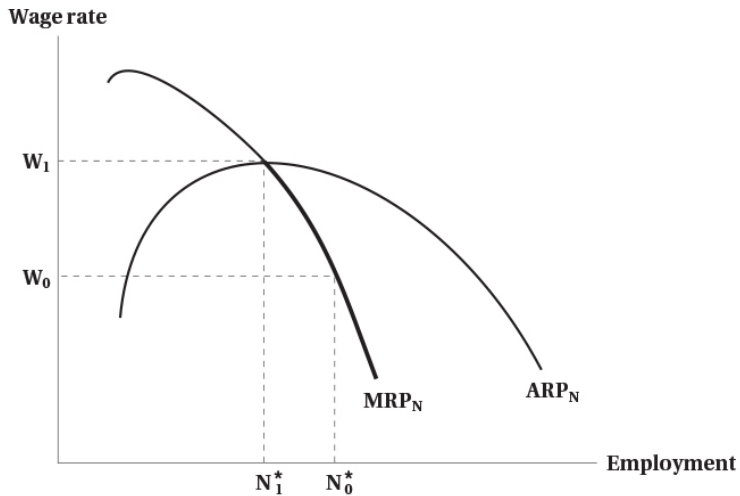
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Figure 5.1: The Firm's Short-Run Demand for Labour



# Shutdown Region and Curve Segment

- Shut down if the wage exceeds the **average revenue product** of labour.
- Short-run labour demand coincides with  $MRP_N$  below the  $ARP-MRP$  intersection.
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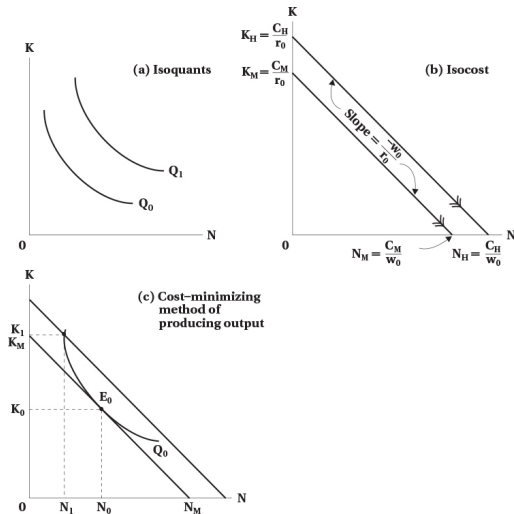
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- **Isoquant:** Combinations of  $K$  and  $N$  yielding a given  $Q$ ; diminishing MRTS.
  - **MRTS** – the rate at which a firm can substitute labor for capital while keeping production output constant
- **Isocost:**  $TC = rK + wN$ , all input bundles with the same cost.

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## Figure 5.2: Isoquants, Isocost, and Cost Minimization



## Panel (a): Isoquants

- Curves labeled  $Q_0$  and  $Q_1$  are **isoquants**
- Each isoquant shows all  $(K, N)$  combinations producing the same output
- $Q_1$  represents a higher level of output than  $Q_0$
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# Isoquants: Interpretation

- Moving **along** an isoquant:
  - Output remains fixed
  - Firm substitutes capital for labour or vice versa
- Moving to a **higher isoquant** (from  $Q_0$  to  $Q_1$ ):
  - Requires more inputs
  - Represents an increase in output
- Technological change (not shown):
  - Inward shift: productivity improvement
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## Panel (b): Isocost Lines

- Isocosts show combinations of  $K$  and  $N$  with the same total cost
- Total cost:

$$C = w_0N + r_0K$$

- $w_0$ : wage rate,  $r_0$ : rental rate of capital
- Lines labeled  $C_M$  and  $C_H$  represent different cost levels

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# Isocost Lines: Slopes and Intercepts

- Vertical intercepts:

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- Horizontal intercepts:

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- Slope of the isocost:

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- Increase in total cost  $C$ :
  - Parallel outward shift of the isocost
- Increase in wage  $w_0$ :
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- Firm wants to produce  $Q_0$  at the lowest possible cost
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# Why Other Points Are Not Optimal

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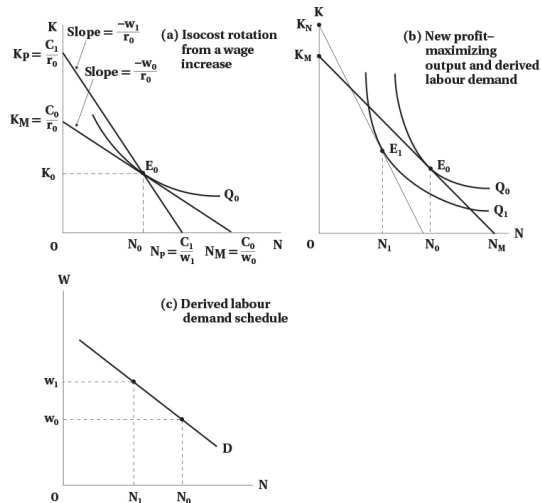
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# Figure 5.3: Deriving the Labour Demand Schedule



# Cost Increases Under Perfect Competition

- Higher  $w$  rotates the isocost inward; substitute  $K$  for  $N$ .
- Firm MC and AC shift up; industry output falls and price rises.

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# Monopoly Case: Cost Increases

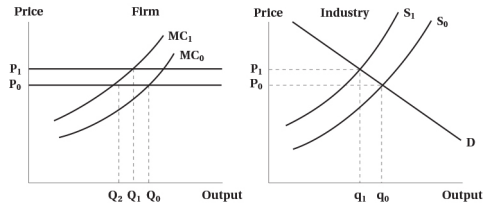
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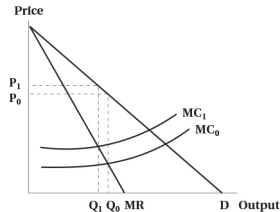
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## Figure 5.4: The Effect of a Cost Increase on Output

(a) Perfect competition



(b) Monopoly



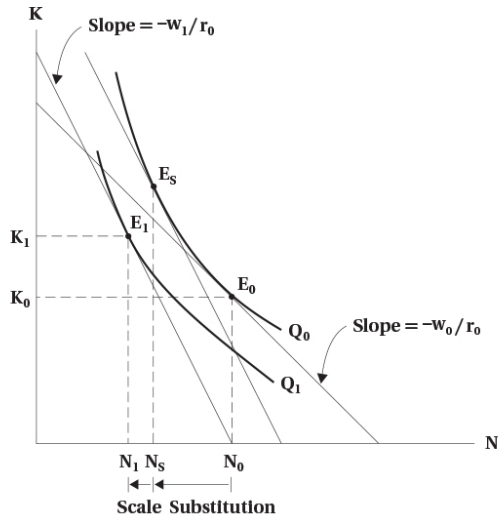
# Substitution vs. Scale Effects

- **Substitution effect:** Use cheaper inputs instead of labour.
- **Scale effect:** Higher costs shrink output scale, reducing all inputs.

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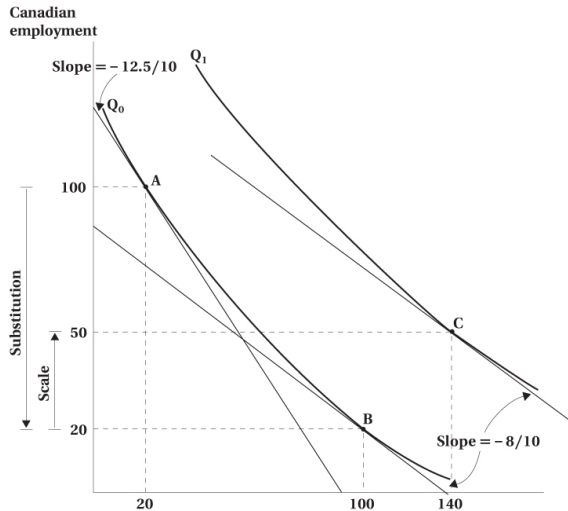
- **Substitution effect:** Use cheaper inputs instead of labour.
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Figure 5.5: Substitution and Scale Effects of a Wage Change



## Worked Example: Offshoring (Toy Inc.)

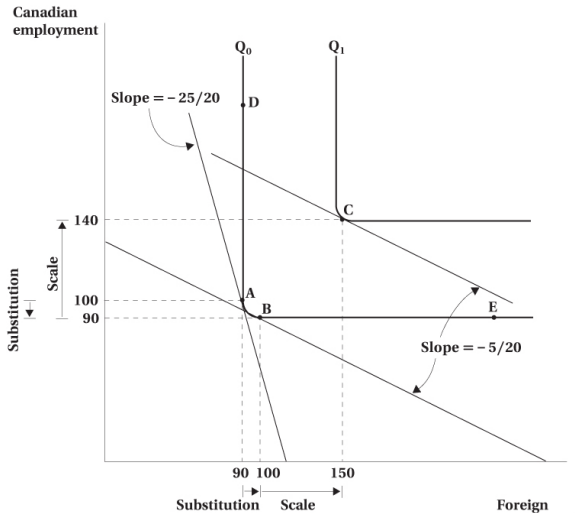
Toy Inc. can hire assembly workers in Canada or abroad; workers are **close substitutes**. Outsourcing abroad  $\Rightarrow$  lower effective wage for that task  $\Rightarrow$  **reduces Canadian employment**.



## Worked Example: Offshoring (Bridge Inc.)

Bridge Inc. builds bridges; Canadian engineers/designers/managers work at home; construction workers hired abroad. Inputs are **complements**. Outsourcing on-site construction **does not reduce** Canadian employment of complementary skilled workers.

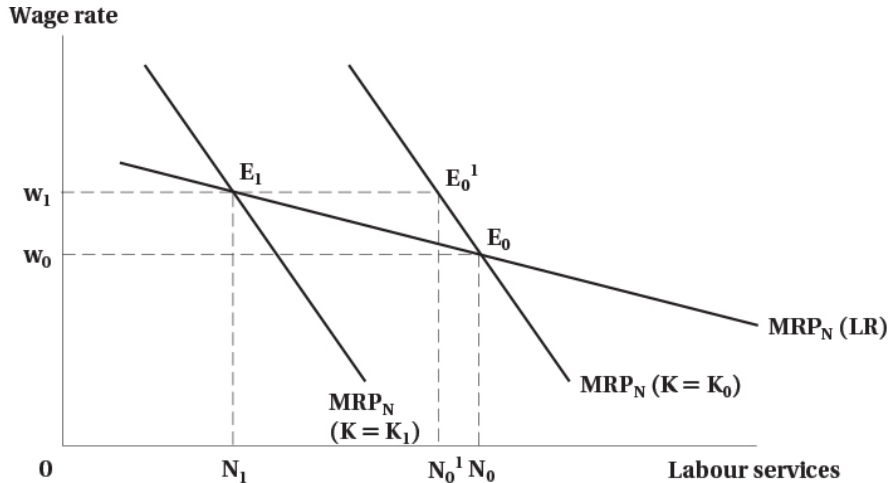
# Bridge Inc. — Illustration



- **Short run:**  $K$  fixed; no substitution effect.
- **Long run:**  $K$  variable; substitution  $\Rightarrow$  larger employment response to wage changes.

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Figure 5.6: Labour Demand in Short vs. Long Run



# Cost Minimization in Public/Quasi-Public Sectors

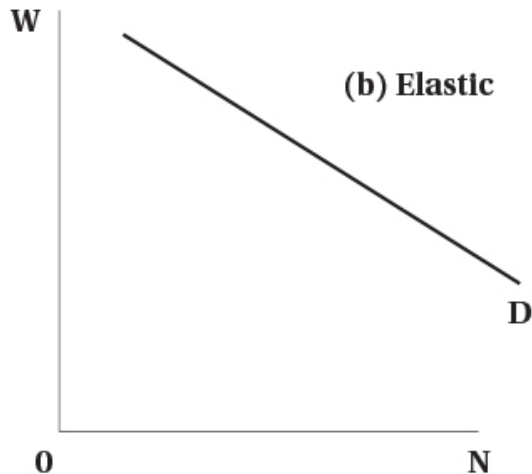
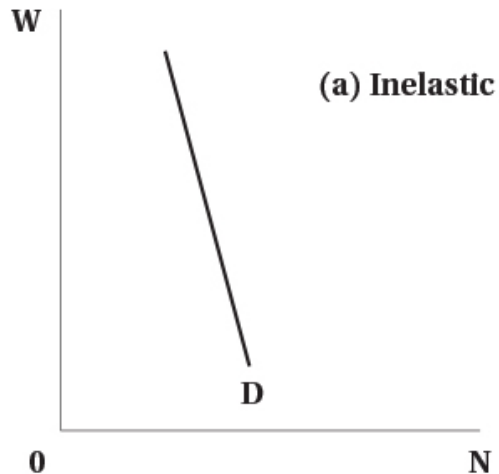
Organizations target **given output at minimum cost**. Scale vs. substitution effects remain key to labour demand responses.

- Demand for labour typically falls when wages rise.
- Elasticity  $\varepsilon_{N,w} = \frac{\% \Delta N}{\% \Delta w}$ .

# Definition and Measurement

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Figure 5.7: Inelastic vs. Elastic Labour Demand



# Determinants of Elasticity

- Availability of **substitute inputs**.
- Supply conditions of substitute inputs.
- Elasticity of demand for **output**.
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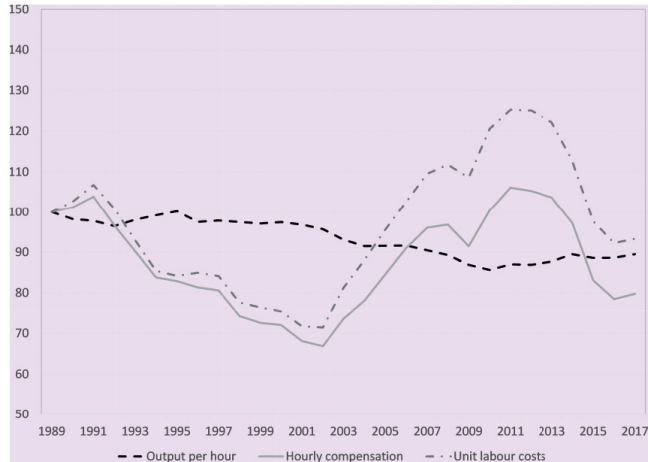
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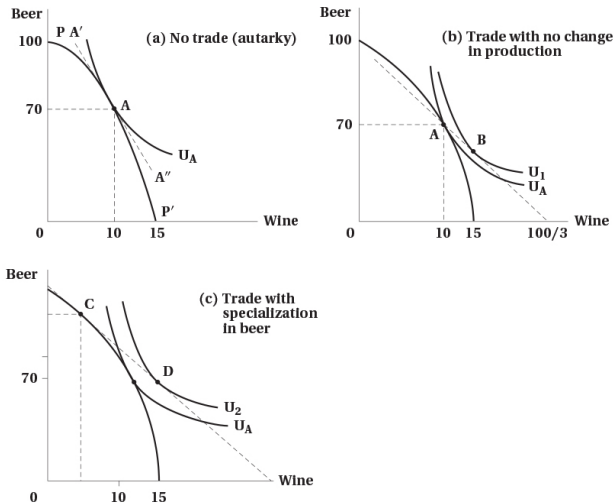
# TABLE 5.1: Productivity (2000) and Changes in Productivity, Hourly Compensation, and Unit Labour Costs, 2000–2018

| Country        | Productivity in 2000 (as % of Canada) | % Change in Productivity | % Change in Compensation |
|----------------|---------------------------------------|--------------------------|--------------------------|
| Korea          | 45                                    | 99                       | 32                       |
| Mexico         | 46                                    | 2                        | –44                      |
| New Zealand    | 79                                    | 21                       | 76                       |
| Japan          | 87                                    | 20                       | –38                      |
| Australia      | 99                                    | 24                       | 45                       |
| Canada         | 100                                   | 18                       | 16                       |
| Spain          | 102                                   | 17                       | 18                       |
| United Kingdom | 111                                   | 18                       | –5                       |
| Finland        | 116                                   | 20                       | 24                       |
| Italy          | 119                                   | 1                        | 14                       |
| Sweden         | 121                                   | 28                       | 17                       |
| United States  | 123                                   | 30                       | 2                        |
| Germany        | 126                                   | 19                       | 9                        |
| France         | 129                                   | 18                       | 19                       |
| Netherlands    | 135                                   | 15                       | 19                       |
| Denmark        | 135                                   | 22                       | 33                       |
| Switzerland    | 136                                   | 19                       | 40                       |
| Belgium        | 143                                   | 14                       | 21                       |
| Norway         | 164                                   | 16                       | 46                       |

Figure 5.8: Relative Trends in Labour Costs and Productivity (Canada vs. U.S.)



# Figure 5.9: Trade, Consumption, and Production in a Simple Economy



- Short-run adjustment costs existed.
- Long-run benefits of FTA were significant.
- Measured impacts of offshoring on employment have been modest.

## Evidence: CUFTA/NAFTA & Offshoring

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